

Fedor V. Gerasimov

Ph.D. Student in Mathematics | Rice University | Houston, TX

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EDUCATION

Ph.D. Student in Mathematics <i>Rice University</i> Advisor: Gregory Chambers. Research focus: discrete, convex, and computational geometry.	2025–present
M.A. in Mathematics <i>Rice University</i>	August 2026
B.Sc. in Applied Mathematics and Computer Science <i>Moscow Institute of Physics and Technology (MIPT)</i> Thesis: <i>Blocking Numbers of Convex Bodies</i> . Advisor: Alexander A. Polyanskii.	2019–2023
Graduate-Level Study in Data Analysis and Machine Learning <i>Yandex School of Data Analysis</i> Advanced coursework in machine learning, algorithms, and applied statistics.	2022–2023

RESEARCH EXPERIENCE

Doctoral Researcher — Discrete, Convex, and Computational Geometry <i>Rice University · Advisor: Gregory Chambers</i>	2025–present
Graduate Researcher — Rank-Deficient Least Squares <i>Rice University · Advisor: Matthias Heinkenschloss</i> Investigated parameter identifiability and local solution manifolds of rank-deficient nonlinear least squares problems, with applications in computational biology.	2023–2025
Bachelor’s Thesis — Blocking Numbers of Convex Bodies <i>MIPT · Advisor: Alexander A. Polyanskii</i> Studied blocking numbers of convex bodies and the Hadwiger’s covering conjecture. Thesis (Russian).	2022–2023
Undergraduate Researcher — Maximum-Sum Matchings <i>MIPT · Advisor: Alexander A. Polyanskii</i> Studied intersections of ellipses induced by maximum-sum matchings of planar point sets.	Summer 2021
Research Practicum — Rigidity of Laman Graphs <i>MIPT · Advisor: Alexander A. Polyanskii</i> Developed a proof of Laman graph rigidity using the theory of places, an algebraic method also used in the proof of the Bellows conjecture.	2020–2021

PUBLICATIONS AND POSTERS

Exploring Local Solution Manifolds of Nonlinear Least Squares Problems F. Gerasimov, M. Heinkenschloss, S. Namazifard, A. Khaddaj, and F. Gabbiani. <i>Poster</i> . Rice University Research Training Group workshop; SIAM Texas–Louisiana Section Annual Meeting. PDF.	2024
On an Interesting Circle in a Triangle A. S. Shutov and F. V. Gerasimov. <i>Moscow Mathematical Conference of High School Students</i> . PDF.	2019

TEACHING AND MENTORING

Teaching Assistant — MATH 221: Honors Calculus III

Fall 2026

Rice University

Hold office hours, lead recitations, and run weekly proof-writing tutorials for groups of three to four students.

Directed Reading Program Organizer — Applications of Linear Algebra

Spring 2026

Rice University

Organized an undergraduate reading group on linear algebra in geometry, algorithms, and optimization.

Mathematics Instructor — Techniques of a Pro Mathematician

Summer 2025

Tapia Camps, Rice University

Taught a week-long course for high-school students on mathematical problem solving and proof strategies.

Teaching Assistant — Optimization and Operations Research

Spring 2025

Rice University · CMOR 360: Introduction to Optimization and Operations Research

Led recitations, held office hours, and created and graded assignments.

Competitive Mathematics Coach

Sept. 2021–May 2022

Lyceum “The Second School”

Taught two two-hour classes per week to prepare students for national and regional mathematics competitions.

SELECTED CONFERENCES

Attendee — Computational Geometry (SoCG 2026)

June 2026

42nd International Symposium on Computational Geometry, New Brunswick, New Jersey.

Poster Presenter — SIAM Texas–Louisiana Section Annual Meeting

October 2024

Baylor University, Waco, Texas. Conference.

Attendee — Geometry and Topology of 3-Manifolds

2022

International conference, Sochi, Russia.

Speaker — Contemporary Problems of Applied Combinatorics

2021

Presented a talk on modeling an epidemic on a random graph. Sochi, Russia.

PROFESSIONAL EXPERIENCE

Python Developer — Interfax Digital Solutions

Oct. 2022–July 2023

Developed graph algorithms based on the analysis of media texts.

ACADEMIC SERVICE

Graduate Student Association Representative

2025–2026

Represented the Department of Mathematics in Rice University’s Graduate Student Association.

SELECTED AWARDS

Best Paper Award — Moscow Mathematical Conference of High School Students

2020

Silver Medal — All-Russian National Mathematics Olympiad

2019

Gold Medals — Higher School of Economics Mathematics Olympiad

2018, 2019

Silver Medals — Moscow Mathematical Olympiad

2018, 2019

SKILLS

Programming: Python, C++, R, MATLAB, Wolfram Mathematica, Lean 4, Gurobi Optimization.

Languages: Russian (native), English (proficient), Spanish (beginner).